



Modelling Financial Markets

Chapter 20 (special chapter): Path Signatures and Realised Volatility — a
Research Case Study



Daniel Traian PELE

Bucharest University of Economic Studies

IDA Institute Digital Assets

Blockchain Research Center

AI4EFin Artificial Intelligence for Energy Finance

Romanian Academy, Institute for Economic Forecasting

MSCA Digital Finance

Acronyms used in this material (1/2)

- ▣ **AAPL**: Apple Inc. (Nasdaq ticker)
- ▣ **ACM**: Association for Computing Machinery
- ▣ **AI**: Artificial Intelligence
- ▣ **AR**: AutoRegressive (model); in event studies: Abnormal Return
- ▣ **ARIMA**: AutoRegressive Integrated Moving Average (model)
- ▣ **AUD**: Australian Dollar (ISO 4217 code)
- ▣ **BH**: Benjamini–Hochberg (procedure)
- ▣ **BIC**: Bayesian Information Criterion
- ▣ **BPQ**: Bollerslev–Patton–Quaedvlieg (insanity filter of the HARQ paper)
- ▣ **CAD**: Canadian Dollar (ISO 4217 code)
- ▣ **CL**: Crude oil (WTI light sweet) futures, NYMEX ticker
- ▣ **COVID-19**: COronaVirus Disease 2019
- ▣ **DM**: Diebold–Mariano (test)
- ▣ **ES**: E-mini S&P 500 futures (CME ticker)
- ▣ **EURUSD**: Euro / US dollar exchange rate (FX pair code)
- ▣ **FDR**: False Discovery Rate
- ▣ **FWER**: Family-Wise Error Rate
- ▣ **FX**: Foreign eXchange
- ▣ **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ **GC**: Gold futures, COMEX ticker
- ▣ **GW**: Giacomini–White (test)
- ▣ **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)

Acronyms used in this material (2/2)

- ▣ **HAR**: Heterogeneous AutoRegressive (model)
- ▣ **HARQ**: HAR model with Realised Quarticity
- ▣ **JPM**: JPMorgan Chase & Co. (NYSE ticker)
- ▣ **KDD**: Knowledge Discovery and Data Mining (ACM SIGKDD conference)
- ▣ **LARS**: Least-Angle RegreSsion (algorithm)
- ▣ **LASSO**: Least Absolute Shrinkage and Selection Operator
- ▣ **MCS**: Model Confidence Set
- ▣ **MSE**: Mean Squared Error
- ▣ **NG**: Natural Gas futures, NYMEX ticker
- ▣ **NVDA**: NVIDIA Corporation (Nasdaq ticker)
- ▣ **OLS**: Ordinary Least Squares
- ▣ **PDE**: Partial Differential Equation
- ▣ **PRDS**: Positive Regression Dependence on a Subset (of p-values)
- ▣ **QLIKE**: Quasi-LIKElihood (loss function)
- ▣ **RQ**: Realised Quarticity
- ▣ **RV**: Realised Variance
- ▣ **SHAR**: Semivariance HAR model
- ▣ **TSLA**: Tesla, Inc. (Nasdaq ticker)
- ▣ **US**: United States
- ▣ **USDJPY**: US dollar / Japanese yen exchange rate (FX pair code)
- ▣ **VOLARE**: VOLatility Archive for Realized Estimates (database)
- ▣ **XOM**: Exxon Mobil Corporation (NYSE ticker)

Chapter Route and Outcomes

Route of this chapter

- A research case study: the method of [Gu et al. \(2024\)](#), published for freight rates at Amazon, transferred to daily realised variance of 50 VOLARE assets [Cipollini et al. \(2026\)](#)
- Part I: realised measures, the data, the HAR benchmarks, path signatures
- Part II: the paper of Gu et al., a design fixed before the results, results, robustness, one pass of the AI-assisted discovery loop
- Implementation detail sits in the [Appendix](#) at the end

After this chapter you can

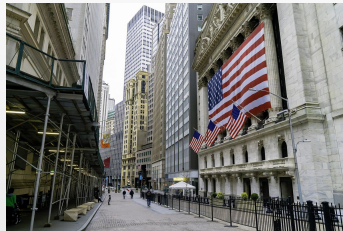
- compute the signature of a piecewise-linear path and read its Lévy area
- build signature features and kernel weights for a volatility forecast
- compare forecasts on many assets with DM tests, model confidence sets and Holm/BH corrections
- pre-register a method transfer and report a null result with its mechanism

Why Forecast Realised Variance?

- Volatility is an input to almost every risk measure in this course
 - ▶ Tail-risk measures ([Chapter 7](#)), option prices ([Chapter 12](#)), portfolio weights ([Chapter 4](#))
- Intraday data make daily variance almost observable: realised variance (RV, [Chapter 9](#))
 - ▶ So volatility forecasting becomes a regression problem with a noisy but model-free target
- The benchmark is hard to beat: the HAR model [Corsi \(2009\)](#) ([Chapter 9](#))
 - ▶ Three averages of past RV; linear; estimated by OLS in a second
- Any new method must beat HAR out of sample, on many assets, with a proper test

The Research Question

- Gu et al. describe a recent window of a series as a **path** and summarise it by its **signature**
 - ▶ Signature features enter a sparse linear regression (two-step LASSO)
 - ▶ Past periods whose path looks like today's get more weight (a signature kernel)
- They report a more than fivefold accuracy gain over an industry forecast of freight rates
 - ▶ Question: does the same recipe improve on HAR-type models for realised variance?
- Two separate ingredients to test: the signature features and the kernel weights



Wall Street and the New York Stock Exchange, 25
March 2020

Photo: [Billie Grace Ward \(2020\)](#); CC BY 2.0; [Wikimedia Commons](#)

Realised Variance and Quadratic Variation (1/2)

- The price model: the log price moves by a drift, a diffusion term and jumps

$$dp_t = \mu_t dt + \sigma_t dW_t + dJ_t$$

- ▶ p_t : log price at time t (time in days); μ_t : drift; σ_t : spot volatility; W_t : Brownian motion; J_t : pure-jump process, $\Delta J_s = \text{jump size at time } s$

- Realised variance: the sum of squared intraday returns of day t [Barndorff-Nielsen & Shephard \(2002\)](#)

$$RV_t = \sum_{i=1}^M r_{t,i}^2 \xrightarrow{M \rightarrow \infty} \int_{t-1}^t \sigma_s^2 ds + \sum_{t-1 < s \leq t} (\Delta J_s)^2$$

- ▶ M : number of intraday returns in day t (for a US stock, 78 five-minute returns in the 6.5-hour session); $r_{t,i}$: the i -th intraday log return of day t
- ▶ The limit is the quadratic variation of day t : the integrated variance plus the sum of squared jumps

Realised Variance and Quadratic Variation (2/2)

- Without jumps and noise, on an equally spaced grid, the limit is the integrated variance; the error is mixed Normal

$$\sqrt{M}(\text{RV}_t - \text{IV}_t) \Rightarrow \text{MN}(0, 2\text{IQ}_t), \quad \text{IV}_t = \int_{t-1}^t \sigma_s^2 ds, \quad \text{IQ}_t = \int_{t-1}^t \sigma_s^4 ds$$

- ▶ IV_t : integrated variance, the true variance of day t ; IQ_t : integrated quarticity; $\Rightarrow$: convergence in distribution
 - ▶ $\text{MN}(0, V)$: mixed Normal, a Normal with a random variance V ; given the volatility path, the error is Normal with variance $2\text{IQ}_t/M$
- Reading: RV is a noisy measurement of IV_t , and the noise is largest on volatile days (where IQ_t is large)
 - ▶ This is the measurement error that HARQ corrects for (next section)
- Microstructure noise makes very fine sampling biased; five-minute returns are the standard compromise ([Chapter 9](#))
- [Chapter 9](#) derives these results; they are used here as given

The Realised Measures in VOLARE (1/2)

- Semivariances Barndorff-Nielsen, Kinnebrock & Shephard (2010): RV split by the sign of the intraday returns

$$RV_t^+ = \sum_{i=1}^M r_{t,i}^2 1\{r_{t,i} > 0\}, \quad RV_t^- = \sum_{i=1}^M r_{t,i}^2 1\{r_{t,i} < 0\}$$

- ▶ $1\{\cdot\}$: indicator, 1 if the condition holds and 0 otherwise; RV_t^+ : variance from up-moves, RV_t^- : variance from down-moves
- ▶ They add up: $RV_t = RV_t^+ + RV_t^-$ (holds in VOLARE to machine precision)
- Realised quarticity: an estimate of IQ_t , the scale of the measurement error of RV

$$RQ_t = \frac{M}{3} \sum_{i=1}^M r_{t,i}^4 \rightarrow IQ_t$$

- ▶ The factor $M/3$: a Normal return with variance s^2 has fourth moment $3s^4$, and each five-minute variance is about IV_t/M
- ▶ The limit needs a continuous, noise-free price and an equally spaced grid; a jump ΔJ adds about $M(\Delta J)^4/3$, so raw RQ is jump-sensitive

The Realised Measures in VOLARE (2/2)

- Bipower variation (robust to jumps), realised kernel (robust to noise) [Barndorff-Nielsen et al. \(2008\)](#), range-based estimators
- Target of this chapter: five-minute RV
 - ▶ Across 400 assets, little beats five-minute RV as a proxy of daily variance [Liu, Patton & Sheppard \(2015\)](#)
 - ▶ Its semivariances and quarticity are on the same grid, so SHAR and HARQ use consistent inputs

Targets and Losses (1/2): the Target

- The target at horizon h is the average RV over the next h days

$$RV_{t+1:t+h} = \frac{1}{h} \sum_{j=1}^h RV_{t+j}, \quad h \in \{1, 5, 22\}$$

- ▶ t : the forecast origin (the last day observed); $h = 1, 5, 22$: one day, one week, one month of trading days

- Log models forecast $y = \ln RV_{t+1:t+h}$ and return to levels with a variance correction

$$F_t = \exp(\hat{y}_t + \hat{s}^2/2)$$

- ▶ F_t : forecast of $RV_{t+1:t+h}$ made at t ; $\hat{y}_t$: fitted log forecast; $\hat{s}^2$: variance of the regression residuals
- ▶ $+\hat{s}^2/2$: since $\mathbb{E}[e^Z] = e^{\mu+s^2/2}$ for $Z \sim N(\mu, s^2)$; exact if the log error is conditionally Normal with constant variance, otherwise an approximation
- For $h > 1$ targets overlap: this induces serial dependence (through lag $h - 1$ by construction, possibly longer), so every test uses a HAC (heteroskedasticity and autocorrelation consistent) variance

Targets and Losses (2/2): Robust Losses

- A loss is **robust** if, for a conditionally unbiased proxy, the *expected*-loss ranking with RV_t equals that with the true variance Patton (2011)
 - ▶ Conditionally unbiased: $\mathbb{E}_{t-1}[RV_t] = \sigma_t^2$, with $\mathbb{E}_{t-1}$ the expectation given information up to $t - 1$ and σ_t^2 the true variance of day t
- The two robust losses used here (written for $h = 1$; for $h > 1$ replace RV_t by the target)

$$MSE_t = (RV_t - F_t)^2, \quad QLIKE_t = \frac{RV_t}{F_t} - \ln \frac{RV_t}{F_t} - 1 \geq 0$$

- ▶ F_t : the forecast of RV_t ; QLIKE is zero only if $F_t = RV_t$ and grows faster when the forecast is too low than when it is too high
- QLIKE depends on the ratio RV_t/F_t : the same proportional error has the same loss on a calm day as on a stormy day
- MSE is dominated by a few crisis days; QLIKE is the primary loss, MSE the secondary one; a model is compared through its mean loss over the evaluation days

VOLARE: an Open Archive of Realised Measures

- Built at the Universities of Messina and Florence from tick data [Cipollini et al. \(2026\)](#)
 - ▶ Asset-specific cleaning, trading calendars and timestamp precision; the same estimators for every asset class
- Free for research at volare.unime.it; not redistributed with this course
 - ▶ Download the realised-variance files and point the notebooks to your local folder
- Citation requested by the authors
 - ▶ Cipollini, Cruciani, Gallo, Insana, Otranto & Spagnolo (2026), arXiv:2602.19732



University of Messina, the home of VOLARE

Photo: [Superchilum \(2013\)](#); CC BY-SA 3.0; [Wikimedia Commons](#)

The Panel: 50 Assets in Three Classes

Class	Assets	Sample	Observations	Median volatility
American large-cap stocks	40	2 Jan 2015 – 30 Jan 2026	111 440	17.1%
Futures (C, CL, ES, GC, NG)	5	28 Sep 2009 – 30 Jan 2026	21 059	19.7%
FX (AUD, EUR, GBP, CAD, JPY vs USD)	5	25 Sep 2009 – 30 Jan 2026	21 203	7.3%

▣ Assets

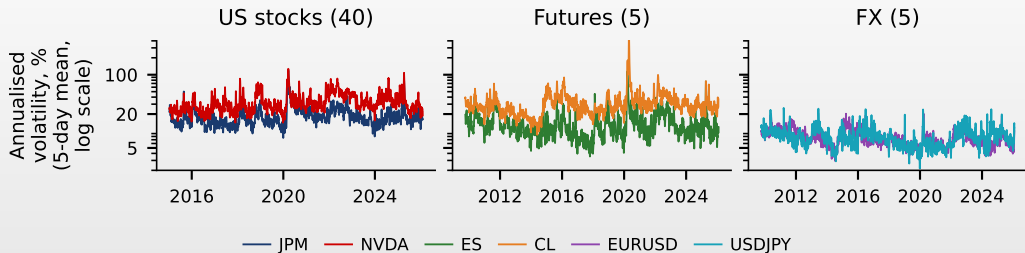
- ▶ Stocks: 40 members of the Dow Jones 30 and S&P 100 (AAPL to XOM)
- ▶ Futures: corn, crude oil, E-mini S&P 500, gold, natural gas

▣ Median volatility: $\sqrt{252 \text{ median}(RV_t)}$ per asset, then the median over the class

- ▶ 252 trading days a year turn a daily variance into an annual one; the data contain trading days only

▣ Variables: five-minute RV, semivariances RV^+ , RV^- , realised quarticity (RQ), daily open and close prices

Realised Volatility Across Asset Classes



- Annualised volatility $\sqrt{252 RV_t}$, five-day mean, log scale, two assets per class (stocks: JPM = JPMorgan Chase, NVDA = Nvidia; futures: ES = E-mini S&P 500, CL = crude oil; FX: EURUSD, USDJPY = US dollar in yen); typical levels: FX about 7.3%, stocks about 17.1% a year
- All classes share March 2020; crude oil (CL) peaks in April 2020, when the front contract traded below zero: a forecaster must adapt within days
- On a log scale volatility is stationary and persistent (the long memory HAR approximates); spikes decay within weeks, so the shape of the recent path carries information

HAR: Three Averages of the Past (1/2)

- HAR (Heterogeneous AutoRegressive) model [Corsi \(2009\)](#): future RV is a linear function of the daily, weekly and monthly averages of past RV

$$RV_{t+1:t+h} = \beta_0 + \beta_d RV_t + \beta_w RV_t^{(w)} + \beta_m RV_t^{(m)} + \varepsilon_{t+h}$$

$$RV_t^{(w)} = \frac{1}{5} \sum_{j=0}^4 RV_{t-j}, \quad RV_t^{(m)} = \frac{1}{22} \sum_{j=0}^{21} RV_{t-j}$$

- Notation

- ▶ $RV_t^{(w)}, RV_t^{(m)}$: weekly (5-day) and monthly (22-day) averages of RV up to day t ; ε_{t+h} : forecast error
- ▶ β_0 : intercept; $\beta_d, \beta_w, \beta_m \geq 0$ in practice: how much yesterday, last week and last month weigh; estimated by OLS for each horizon h
- ▶ $\beta_d + \beta_w + \beta_m$ close to 1: high persistence of volatility

- Traders at daily, weekly and monthly frequencies: a cascade that mimics long memory with three parameters

HAR: Three Averages of the Past (2/2)

- HAR is a restricted AR(22): the 22 lag coefficients take only three distinct values

$$\phi_1 = \beta_d + \frac{\beta_w}{5} + \frac{\beta_m}{22}, \quad \phi_{2..5} = \frac{\beta_w}{5} + \frac{\beta_m}{22}, \quad \phi_{6..22} = \frac{\beta_m}{22}$$

- ▶ ϕ_j : the coefficient of RV_{t+1-j} in an AR(22) (autoregression with 22 lags); writing out the two averages gives these values

- log-HAR: the same regression in logs

$$\ln RV_{t+1:t+h} = \beta_0 + \beta_d \ln RV_t + \beta_w \ln RV_t^{(w)} + \beta_m \ln RV_t^{(m)} + \varepsilon_{t+h}$$

- ▶ No negative forecasts, less driven by spikes; back to levels with $F_t = \exp(\hat{y}_t + \hat{s}^2/2)$
- ▶ Logs of averages are not averages of logs, so this log-HAR is not an exact restricted AR(22) in $\ln RV$ (Seminar 20, A5)

Two Refinements (1/2): HARQ

- RV is a generated regressor: it measures the true variance with an error

$$RV_t = IV_t + \eta_t, \quad \text{Var}(\eta_t) \approx 2 IQ_t / M$$

- η_t : measurement error; errors-in-variables attenuate (pull towards zero) β_d , most on noisy days

- HARQ Bollerslev, Patton & Quaedvlieg (2016): the weight on yesterday's RV falls when RV is measured less precisely

$$RV_{t+1:t+h} = \beta_0 + (\beta_d + \beta_Q(RQ_t^{1/2} - \overline{RQ^{1/2}}))RV_t + \beta_w RV_t^{(w)} + \beta_m RV_t^{(m)} + \varepsilon_{t+h}$$

- $RQ_t^{1/2}$: square root of realised quarticity, a measure of the error of RV_t ; $\overline{RQ^{1/2}}$: its in-sample mean, so β_d is the weight on an average day
- Expected $\beta_Q < 0$: on a noisy day (RQ_t above average) yesterday's RV gets less weight

- BPQ filter: a forecast outside the in-sample range of the target is replaced by the in-sample mean ("insanity filter")

Two Refinements (2/2): SHAR

- SHAR Patton & Sheppard (2015): split the daily term by the sign of the intraday returns

$$RV_{t+1:t+h} = \beta_0 + \beta^+ RV_t^+ + \beta^- RV_t^- + \beta_w RV_t^{(w)} + \beta_m RV_t^{(m)} + \varepsilon_{t+h}$$

- ▶ RV_t^+, RV_t^- : the positive and negative semivariances of day t ; β^+, β^- : their weights in the forecast
- ▶ $\beta^+ = \beta^-$ gives back HAR, since $RV_t^+ + RV_t^- = RV_t$
- Patton and Sheppard find $\beta^- > \beta^+$
 - ▶ Variance from down-moves predicts more future variance than variance from up-moves: an intraday version of the leverage effect (Chapter 5)
- The signature can capture a related effect from daily data: the Lévy area between returns and log RV (Path Signatures section)

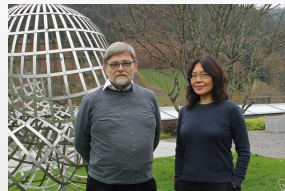
Seven Models in One Table

Model	Target	Weights	Regressors
HAR	RV	equal	$RV_t, RV_t^{(w)}, RV_t^{(m)}$
log-HAR	$\ln RV$	equal	the same, in logs
HARQ	RV	equal	$HAR + RV_t (RQ_t^{1/2} - \overline{RQ^{1/2}})$
SHAR	RV	equal	$RV_t^+, RV_t^-, RV_t^{(w)}, RV_t^{(m)}$
log-HAR-K	$\ln RV$	signature kernel	log-HAR regressors
Sig-L	$\ln RV$	equal	log-HAR + 39 signature terms, two-step LASSO
Sig-LK	$\ln RV$	signature kernel	as Sig-L: the method of Gu et al.

- The last four form a 2×2 ablation (each ingredient switched on and off separately): signature features (yes/no) $\times$ kernel weights (yes/no)
- Log-HAR is the natural baseline for the signature models: same target, and its regressors are among the candidates; but the LASSO may also drop log-HAR terms, so Sig-L vs log-HAR compares two procedures, not only two feature sets

Where Signatures Come From

- **Chen (1957)**: iterated integrals of a path form an algebra and determine it up to reparametrisation
 - ▶ Kuo-Tsai Chen, geometry of loop spaces
- **Lyons (1998)**: rough path theory; the signature is the natural feature set for controlled differential equations
 - ▶ A response to a path depends on it (to first order) only through its signature
- **Machine learning**: [Levin, Lyons & Ni \(2013\)](#), [Chevyrev & Kormilitzin \(2016\)](#), [Lyons & McLeod \(2022\)](#); finance: signature-based asset price models [Cuchiero, Gazzani & Svaluto-Ferro \(2023\)](#)



Terry Lyons (left), Oberwolfach, 2024

Photo: [Katrin Schmid \(2024\)](#); CC BY-SA 2.0 DE; [Wikimedia Commons](#)

A Window of Data Is a Path

- Take the last L days of $d - 1$ series, add time as a channel, and connect the points linearly

$$X : [0, 1] \rightarrow \mathbb{R}^d, \quad X_{u_i} = \left(u_i, \ln \text{RV}_{t-L+1+i}, \sum_{j \leq t-L+1+i} r_j \right), \quad u_i = \frac{i}{L-1}, \quad i = 0, \dots, L-1$$

- ▶ L : window length in days (here 22); d : number of channels (here $d = 3$: time, log RV, cumulative return); $u_i \in [0, 1]$: rescaled time of the i -th point; r_j : open-to-close return of day j
- **Time augmentation**: the first channel is time itself
 - ▶ Without it the signature cannot tell a fast move from a slow one (reparametrisation invariance)
- A regression on the window sees 22 lags; a path summary sees their order, speed and co-movement
 - ▶ Example: RV rose while prices fell, or RV rose first and prices fell later

The Signature: Definition (1/2)

- One signature term is an iterated integral of the path along a word of channels

$$S(X)_{a,b}^{i_1 \dots i_k} = \int_{a < u_1 < \dots < u_k < b} dX_{u_1}^{i_1} \dots dX_{u_k}^{i_k}$$

- Notation

- ▶ $X : [a, b] \rightarrow \mathbb{R}^d$: a path of bounded variation (finite total length), e.g. piecewise linear; X_u^i : its channel i at time u
- ▶ Word $(i_1, \dots, i_k)$: an ordered list of k channels, letters in $\{1, \dots, d\}$; k is the **level** of the term
- ▶ $u_1 < \dots < u_k$: the integral runs over ordered times; dX_u^i : increment of channel i at time u (for a linear segment, slope times du)

The Signature: Definition (2/2)

- The signature is the whole sequence of levels

$$\text{Sig}(X) = (1, S^{(1)}, S^{(2)}, S^{(3)}, \dots), \quad S^{(k)} \in (\mathbb{R}^d)^{\otimes k}$$

- ▶ $S^{(k)}$: the d^k terms of level k , one for each word of length k , stored as a k -dimensional array (a tensor; $\otimes$: tensor product)
- ▶ $\text{Sig}^N(X)$: the truncated signature, levels 0 to N ; N is the depth

- Level 1: the total increments of the channels

$$S^i = X_b^i - X_a^i$$

- Level 2: increments of channel j weighted by how far channel i has already moved

$$S^{ij} = \int_a^b (X_u^i - X_a^i) dX_u^j$$

- ▶ Level 3 adds one more integral; higher levels record finer details of the order of the moves

Linear Segments and Chen's Identity (1/2)

- A linear segment with increment $\Delta \in \mathbb{R}^d$ has the tensor exponential as signature

$$\text{Sig}(\text{segment}) = \exp_{\otimes}(\Delta) = \left(1, \Delta, \frac{1}{2!}\Delta^{\otimes 2}, \frac{1}{3!}\Delta^{\otimes 3}, \dots\right)$$

- ▶ $\Delta = X_b - X_a$: the increment of the segment; $\Delta^{\otimes k}$: the array of all products $\Delta^{i_1} \dots \Delta^{i_k}$, so a level- k term is $\Delta^{i_1} \dots \Delta^{i_k} / k!$

- **Chen's identity** [Chen \(1957\)](#): the signature of two paths run one after the other is the tensor product of their signatures

$$\text{Sig}(X * Y) = \text{Sig}(X) \otimes \text{Sig}(Y), \quad (A \otimes B)^{(k)} = \sum_{p=0}^k A^{(k-p)} \otimes B^{(p)}$$

- ▶ $X * Y$: concatenation, Y starts where X ends; $A^{(k)}, B^{(k)}$: level k of the signatures A, B , with $A^{(0)} = B^{(0)} = 1$
- ▶ Level k of the whole path collects every way of splitting a word of length k between the first and the second piece

Linear Segments and Chen's Identity (2/2)

- ▣ A piecewise-linear path with m segments, increments $\Delta_1, \dots, \Delta_m$

$$\text{Sig} = \exp_{\otimes}(\Delta_1) \otimes \dots \otimes \exp_{\otimes}(\Delta_m)$$

- ▶ This is the whole algorithm: m updates, each costing $O(d^N)$ at depth N (a few lines of numpy); $O(\cdot)$: order of the number of operations

- ▣ Rolling windows: Chen also removes an initial piece

$$\text{Sig}(X_{[s,t]}) = \text{Sig}(X_{[a,s]})^{-1} \otimes \text{Sig}(X_{[a,t]})$$

- ▶ $X_{[s,t]}$: the path restricted to $[s, t]$, with $a \leq s \leq t$; $(\cdot)^{-1}$: inverse for $\otimes$, the signature of the piece run backwards

Worked Example: a Two-Segment Path

Path

- ▣ $X : (0, 0) \rightarrow (1, 0) \rightarrow (1, 2); \Delta_1 = (1, 0), \Delta_2 = (0, 2)$
- ▣ Level 1: $S^{(1)} = \Delta_1 + \Delta_2 = (1, 2)$
- ▣ Level 2 by Chen: $S^{ij} = \frac{1}{2}\Delta_1^i\Delta_1^j + \Delta_1^i\Delta_2^j + \frac{1}{2}\Delta_2^i\Delta_2^j$
- ▣ Δ_m^i : component i of the increment of segment m ; the middle term: channel i moves in segment 1 before channel j moves in segment 2

Result

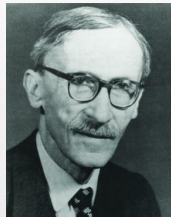
- ▣ $S^{11} = \frac{1}{2}, S^{12} = 0 + 1 \cdot 2 + 0 = 2$
- ▣ $S^{21} = 0, S^{22} = \frac{1}{2} \cdot 2^2 = 2$
- ▣ Check: $S^{11} = (\Delta X^1)^2/2$ and $S^{22} = (\Delta X^2)^2/2$ (shuffle identity)
- ▣ $S^{12} \neq S^{21}$: the order matters (first right, then up)

The Lévy Area: Order as a Number

- The Lévy area A^{ij} : the antisymmetric part of level 2

$$A^{ij} = \frac{1}{2}(S^{ij} - S^{ji})$$

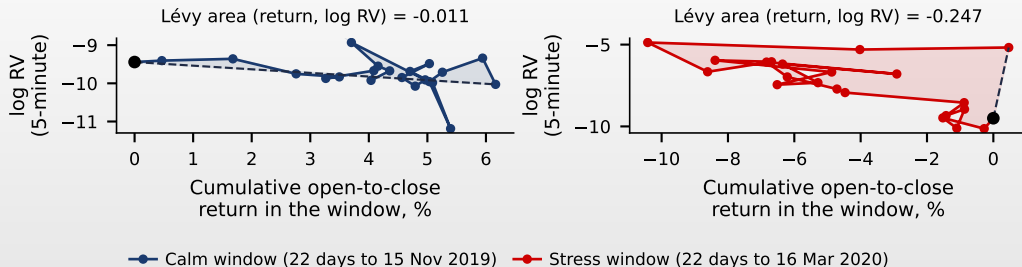
- ▶ Equals the signed area between the path, projected on channels (i, j) , and its chord (the straight line from start to end)
- ▶ $A^{ij} > 0$: channel i tends to move before channel j ; $A^{ij} = -A^{ji}$
- Symmetric part: $\frac{1}{2}(S^{ij} + S^{ji}) = \frac{1}{2} \Delta X^i \Delta X^j$, no new information
 - ▶ Example above: $A^{12} = (2 - 0)/2 = 1$, the area of the triangle under the path
- Channels (return, log RV): a price fall followed by a volatility rise gives $A < 0$, but so does a rise followed by a volatility fall
 - ▶ A lead-lag statistic: leverage-type sequences (Chapter 5) are one contributor to a negative area; the sign alone does not identify them



Paul Lévy (1886–1971)

Photo: Konrad Jacobs; CC BY-SA 2.0 DE;
[Wikimedia Commons](#)

Two Windows of JPM as Paths



- Path (cumulative open-to-close return, log RV) over 22 days; dashed chord from start to end; the shaded region gives the Lévy area
- Calm window (to 15 Nov 2019): log RV moves by -0.58, the path wanders, Lévy area -0.011; stress window (to 16 Mar 2020): log RV rises by +4.32 ($RV \times 75$) while returns fall, Lévy area -0.247
- The net intraday return is only +0.5%: the crash was largely overnight, outside this channel; no signature level recovers what the input omits

Why Signatures Work (1/2): Uniqueness and Shuffles

- **Uniqueness** Hambly & Lyons (2010): the full signature identifies the path
 - ▶ $\text{Sig}(X)$ determines X up to tree-like equivalence (adding pieces that are retraced backwards) and translation
 - ▶ With a strictly increasing time channel nothing can be retraced, so only translation remains
 - ▶ A finite truncation Sig^N is not injective: two different paths can share levels $1..N$
- **Shuffle identity**: a product of signature terms is a sum of higher terms

$$S^i S^j = S^{ij} + S^{ji}, \quad S^{ij} S^k = S^{ijk} + S^{ikj} + S^{kij}$$

- ▶ The sum runs over all interleavings (shuffles) of the two words that keep the order inside each word
- ▶ So a linear model in $\text{Sig}^N(X)$ already contains polynomials of the increments

Why Signatures Work (2/2): Universality and Decay

- **Universality:** any continuous function of the path is approximately linear in the signature (Stone–Weierstrass, via the shuffle identity)

$$\forall \varepsilon > 0 \quad \exists \ell \text{ linear} : \sup_{X \in K} |f(X) - \langle \ell, \text{Sig}(X) \rangle| < \varepsilon$$

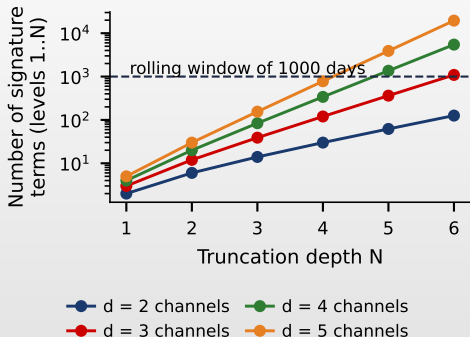
- ▶ K : a compact set of time-augmented paths with a common start (in the 1-variation topology); $f : K \rightarrow \mathbb{R}$: a continuous function of the path
- ▶ $\langle \ell, \text{Sig}(X) \rangle$: a weighted sum of finitely many signature terms, i.e. a linear regression on them
- ▶ This is why a LASSO on signatures can approximate nonlinear functions of the path; the price is d^N features, so sparsity is needed

- **Factorial decay:** high levels are small

$$\|S^{(k)}\| \leq \frac{\|X\|_{\mathbf{1-var}}^k}{k!}$$

- ▶ $\|X\|_{\mathbf{1-var}}$: total length of the path (sum of the lengths of its segments); $\|S^{(k)}\|$: size (Euclidean norm) of level k
- ▶ Truncating at a small depth N loses little for paths of moderate length, which is why channels are rescaled ([▶ Appendix](#))

How Many Features? The Truncation Dimension



- Number of terms in levels 1..N of a d -dimensional path (a geometric sum, since level k has d^k words)
 - ▶ $\dim(d, N) = \sum_{k=1}^N d^k = \frac{d^{N+1}-d}{d-1}$
- This chapter: $d = 3$ channels (time, log RV, cumulative return), depth $N = 3$: $3 + 9 + 27 = 39$ terms
 - ▶ Three are constant (words with only the time letter), so 36 vary
- $d = 5$, $N = 4$ would give 780 terms for 1000 observations: the LASSO would select from noise
- Gu et al. do not disclose N and L ; here $N = 3$, $L = 22$ are fixed ex ante, with $N = 2$ as a robustness check

Two Invariances to Keep in Mind

- **Translation invariance:** $\text{Sig}(X + c) = \text{Sig}(X)$ for any constant vector $c \in \mathbb{R}^d$: only increments enter
 - ▶ The signature does not see the **level** of volatility, only the shape of its recent path
 - ▶ So the level must enter separately: the log-HAR terms play the role of the external factors x_τ of Gu et al.
- **Reparametrisation invariance:** running the path faster or slower, $X_{\psi(u)}$ with ψ increasing, leaves Sig unchanged
 - ▶ Removed by the time channel; basepoint augmentation (prepending a segment from 0 to the original starting point) would remove the first invariance [Morrill et al. \(2020\)](#)
 - ▶ Translating the path to start at 0 does not: it is itself a translation (Seminar 20, C1 uses the prepended segment)

The Signature Kernel

- Kernel = inner product of feature maps [Király & Oberhauser \(2019\)](#): a similarity score between two paths

$$k_N(X, Y) = \langle \text{Sig}^N(X), \text{Sig}^N(Y) \rangle = 1 + \sum_{k=1}^N \sum_{|w|=k} S(X)^w S(Y)^w$$

- ▶ X, Y : two paths with the same channels; w : a word, $|w|$: its length (level); $S(X)^w$: the term of word w ; the leading 1 is level 0
- Distance induced by the kernel (Gu et al., Eq. 9): squared Euclidean distance between the truncated signatures

$$d_{\text{Sig}}(X, Y) = k_N(X, X) - 2k_N(X, Y) + k_N(Y, Y) = \|\text{Sig}^N(X) - \text{Sig}^N(Y)\|^2$$

- ▶ $d_{\text{Sig}} = 0$: the two windows have the same signature up to depth N ; large d_{Sig} : different shapes
- Gu et al. and this chapter use this **truncated** kernel, with N as for the features
 - ▶ The untruncated kernel solves a Goursat PDE (partial differential equation) [Salvi et al. \(2021\)](#); the paper cites it but does not use it
- Channels are rescaled before the kernel (▶ [Appendix](#)): otherwise the channel with the largest units dominates

The Problem at Amazon and the Features

- Weekly North American truckload rates, national and five regions, 2018–2022, forecast up to 12 weeks ahead [Gu et al. \(2024\)](#)
 - ▶ More than 100 candidate factors screened to about 40; in Amazon operations since November 2022; KDD 2024 (ACM Conference on Knowledge Discovery and Data Mining)
- Features at week τ (Section 4.1, Eq. 14): external factors x_τ and the signature of the time-augmented target path
 - ▶ $\xi_\tau = [x_\tau, \text{Sig}^N(\hat{y}_{\tau-l:\tau})]$, $\hat{y}_{\tau-l:\tau} = \{(t_i, y_{t_i})\}$
 - ▶ y_{t_i} : the rate in week t_i , $\tau - l \leq t_i \leq \tau$; l : lookback in weeks;
 $\hat{y}_{\tau-l:\tau}$: the time-augmented path of the last l weeks
- One direct regression per horizon Δt (in weeks), $y_{\tau+\Delta t}$ on ξ_τ ; lookback l , depth N , temperature γ (next slides) are not published (confidentiality)



A container truck on an American interstate, 2025

Photo: [DanTD \(2025\)](#); CC BY 4.0; [Wikimedia Commons](#)

The Method: Two-Step LASSO

- Step 1, selection by a weighted LASSO (Least Absolute Shrinkage and Selection Operator) [Tibshirani \(1996\)](#) (Eq. 14)

$$\hat{\theta}_L = \arg \min_{\theta} \sum_{\tau} w_{\tau} (y_{\tau+\Delta t} - \xi_{\tau}^{\top} \theta)^2 + \lambda \|\theta\|_1$$

- θ : coefficient vector; $w_{\tau} \geq 0$: weight of week τ (equal, or from the kernel, next slide); $\|\theta\|_1 = \sum_j |\theta_j|$
- $\lambda \geq 0$: penalty; a larger λ sets more coefficients exactly to zero; $\lambda = 0$ is weighted OLS

- Step 2, refit by weighted OLS on the selected support (Eq. 15)

$$\hat{\theta} = \arg \min_{\text{supp}(\theta) = \text{supp}(\hat{\theta}_L)} \sum_{\tau} w_{\tau} (y_{\tau+\Delta t} - \xi_{\tau}^{\top} \theta)^2$$

- $\text{supp}(\theta)$: the support, the set of nonzero coefficients; only the regressors kept in step 1 are re-estimated

- Why refit: LASSO shrinks the kept coefficients towards zero; OLS after selection removes that bias [Belloni & Chernozhukov \(2013\)](#)

- The paper calls it “adaptive” because of the weights w_{τ} , not in the sense of the adaptive LASSO of [Zou \(2006\)](#)

- How λ is chosen is not reported

The Method: Kernel Weights

- Distance of each past window to the last window with an observed target (Eq. 11)

$$\delta_{\tau} = d_{\text{Sig}, N}(z_{\tau-l:\tau}, z_{t-\Delta t-l:t-\Delta t})$$

- $z_{\tau} = (x_{\tau}, y_{\tau+\Delta t})$: the *pairs* of predictors and later target, not the features ξ_{τ} (Algorithm 1); $z_{\tau-l:\tau}$: their path over l weeks
- t : the current week; $t - \Delta t$: the last week whose target y_t is already observed; $d_{\text{Sig}, N}$: the signature distance at depth N

- Softmax weights with temperature $\gamma \geq 0$ (Eq. 12): similar windows (small δ_{τ}) get more weight

$$w_{\tau} = \frac{\exp(-\gamma \delta_{\tau})}{\sum_s \exp(-\gamma \delta_s)}$$

- s : runs over all training weeks, so the weights sum to 1; $\gamma = 0$: ordinary (equal-weight) regression; $\gamma \rightarrow \infty$: only the most similar past window
- A nearest-neighbour idea inside a regression: learn from the past periods that looked like today
 - Cost of the idea: fewer effective observations, $n_{\text{eff}} = 1 / \sum_{\tau} w_{\tau}^2$ (Kish); $n_{\text{eff}} = n$ for equal weights $1/n$, $n_{\text{eff}} = 1$ if one week takes all the weight

What the Paper Reports, Read Critically

- National rates, three months ahead, April–November 2021 (Table 1): relative error $|y - \hat{y}|/|y|$ (y : realised rate, $\hat{y}$: its forecast) of about 1–2% vs 13–27% for the industry forecast, “more than fivefold”
 - ▶ Horizons 1–12 weeks (Table 2): 1.06–2.60% nationally, 1.05–5.69% by region; estimated annual saving of about 50 million dollars
 - ▶ Kernel ablation (Table 3, six weekly values, 31 October–5 December 2021, one region): errors up to 9.85% without the kernel vs up to 5.15% with it
- Limits of the evidence
 - ▶ **One benchmark:** an unnamed commercial forecast; no ARIMA, no LASSO without signatures, no random walk
 - ▶ **Short evaluation:** a few months, a few series, no standard errors, no test of equal accuracy
 - ▶ **Undisclosed tuning:** I , N , γ , λ and how they were chosen
- None of this makes the method wrong; it makes the evidence thin, which is why it is tested elsewhere, replication before extension ([Chapter 19](#))

What Transfers to Volatility, and What Does Not

Transfers

- ▣ Signature of the time-augmented recent path of the target
- ▣ Two-step LASSO with weights, direct regression per horizon
- ▣ Softmax weights $\exp(-\gamma \delta_\tau)$ with the truncated signature kernel

Must be adapted

- ▣ Daily, noisy target instead of weekly rates: logs, HAC inference, robust losses
- ▣ External factors x_τ replaced by the log-HAR terms (they carry the level)
- ▣ Undisclosed choices (l, N, γ, λ) fixed ex ante in this chapter
- ▣ Kernel: the path of this chapter (time, ln RV, return) ending at τ vs the window ending at t , updated every 5 days, instead of the pairs $(x_\tau, y_{\tau+\Delta t})$ ending at $t - \Delta t$: an adaptation
- ▣ Strong benchmarks and formal tests instead of one commercial forecast

Hypotheses

- ▣ **H1 (features):** signature terms add information beyond log-HAR
 - ▶ Test: Sig-L vs log-HAR, same target, same equal weights
- ▣ **H2 (weights):** kernel weights improve a given regression
 - ▶ Tests: Sig-LK vs Sig-L and log-HAR-K vs log-HAR
- ▣ **H3 (method):** the full method of Gu et al. beats the HAR family
 - ▶ Tests: Sig-LK vs HAR and vs log-HAR; model confidence set of all seven models
- ▣ **H4 (stress):** any gain is larger in stress periods (COVID-19 crash, 2–30 April 2020, the US tariff shock), where similar past windows should matter most

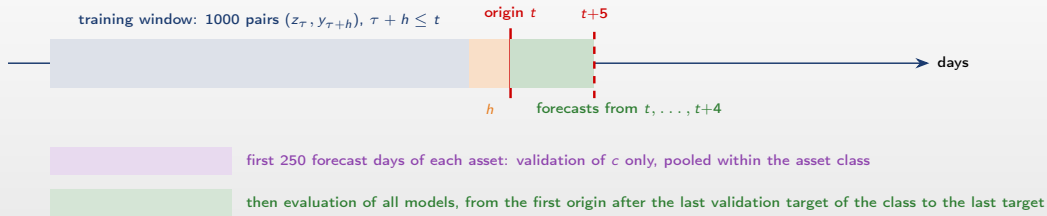
Kernel Temperature: One Choice per Class, on a Validation Block

- Scale-free temperature, set in every training window

$$\gamma = \frac{c}{\text{median}_{\tau}(\delta_{\tau})}$$

- ▶ δ_{τ} : kernel distances of the training days to the current window; $c > 0$: a dimensionless constant, the only tuning parameter
- ▶ Dividing by the median removes the units of δ_{τ} ; $c = 1$: a window at the median distance gets weight e^{-1} relative to an identical one; larger c : more concentrated weights
- Grid $c \in \{0.5, 1, 2, 4\}$, chosen per horizon and asset class on the first 250 forecast days of each asset, pooled within the class
 - ▶ Criterion: mean over the class of $\text{QLIKE}(c)$ divided by the asset's average over the grid
 - ▶ Evaluation starts after the last validation target of the class, so no tuning target postdates an evaluated origin
- For the 40 stocks the smallest temperature, $c = 0.5$ (the most nearly equal weights), wins at every horizon for Sig-LK; for futures and FX the choice changes with the horizon: a flat, noisy criterion (all values in the [Appendix](#))

Rolling Estimation, Step by Step



- Re-estimate every 5 trading days on the last 1000 complete pairs; between re-estimations the coefficients are fixed but the regressors are updated daily
- Kernel distances are recomputed at each re-estimation, against the window ending at the origin t

Evaluation Protocol (1/2): the DM Test

- Per asset and horizon: mean QLIKE and MSE, compared by the DM (Diebold–Mariano) test of equal predictive accuracy [Diebold & Mariano \(1995\)](#)

$$d_t = L(RV_t, F_t^A) - L(RV_t, F_t^B), \quad \text{DM} = \frac{\bar{d}}{\sqrt{\hat{\Omega}/T}}$$

- ▶ L : the loss (QLIKE or MSE); F_t^A, F_t^B : forecasts of models A and B ; d_t : loss differential; $\bar{d}$: its mean over the T evaluation days
- ▶ $\hat{\Omega}$: Newey–West (HAC) long-run variance of d_t , bandwidth $\max(h, \lceil T^{1/3} \rceil)$ lags; $\lceil \cdot \rceil$: rounding up
- Reading: under H_0 : $\mathbb{E}[d_t] = 0$, DM is approximately $N(0, 1)$
 - ▶ $\text{DM} < -1.645$: model A significantly more accurate at 5% (one-sided); $\text{DM} > 1.645$: significantly less accurate
 - ▶ Rolling estimation, so the test is read as the unconditional GW (Giacomini–White) test; the conditional GW test also asks whether d_{t-h} predicts d_t (instruments $(1, d_{t-h})$) [Giacomini & White \(2006\)](#)

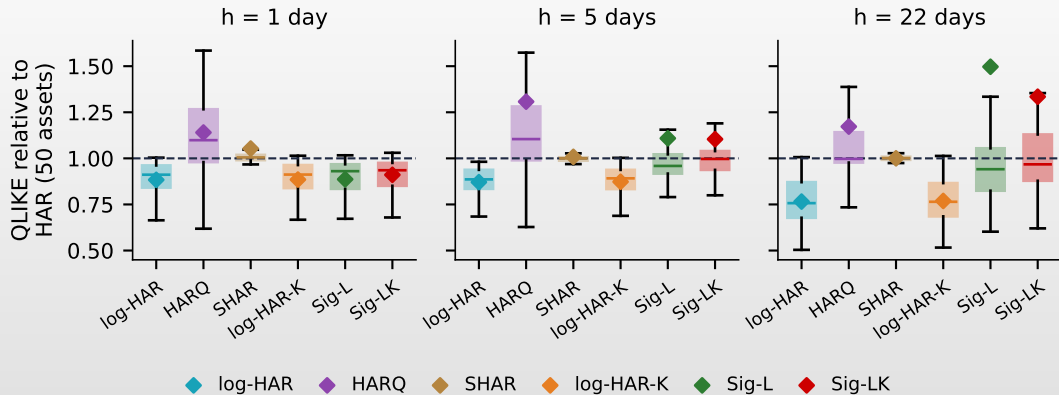
Evaluation Protocol (2/2): Many Models, Many Assets

- MCS (model confidence set) at 90% [Hansen, Lunde & Nason \(2011\)](#): the set of models that contains the best one with probability at least 90%
 - ▶ Built by elimination: while the $T_{\max}$ statistic (the largest standardised gap between a model's mean loss and the average over the models still in the set) rejects equal accuracy, drop the worst model
 - ▶ p -values by the stationary bootstrap, 1000 replications, mean block length $\max(h, 10)$ days
- Panel: DM test on the cross-sectional mean loss differential (one series per date)
 - ▶ Counting wins over 50 assets: the Holm correction controls the FWER (family-wise error rate, the probability of at least one false rejection), BH [Benjamini & Hochberg \(1995\)](#) the FDR (false discovery rate, the expected share of false rejections among the rejections)
- Subperiods: 20 Feb–30 Apr 2020, 2–30 Apr 2025, all other days; the paper's relative error as a descriptive statistic
- 265 929 out-of-sample forecasts per model; stocks are evaluated from January 2020 (the COVID-19 crash is in every stock sample), futures and FX from 2014 (dates in the [Appendix](#))

A Disclosed Deviation

- Pre-registered: the BPQ insanity filter on all models
 - ▶ A test run of the code on JPM (one asset, before any panel result) showed it replacing correct crash forecasts by the calm in-sample mean
 - ▶ On 12 March 2020 the realised next-day RV of JPM was about 40 times the filtered HAR forecast
- Changed before estimating the models on the 50-asset panel: primary results with positivity only (a forecast ≤ 0 becomes the in-sample mean)
 - ▶ The BPQ filter is kept as a robustness check and reported in full
- Rule: a deviation is allowed, a hidden deviation is not; state what was seen when it was changed

QLIKE Relative to HAR, Asset by Asset (1/3)



QLIKE Relative to HAR, Asset by Asset (2/3)

- Ratio for each asset: mean QLIKE of the model over mean QLIKE of HAR; below 1: more accurate than HAR
 - ▶ Boxes: interquartile range over the 50 assets; line: median; diamond: mean; whiskers: 1.5 interquartile ranges

 MFM_ch20_case_study

QLIKE Relative to HAR, Asset by Asset (3/3)

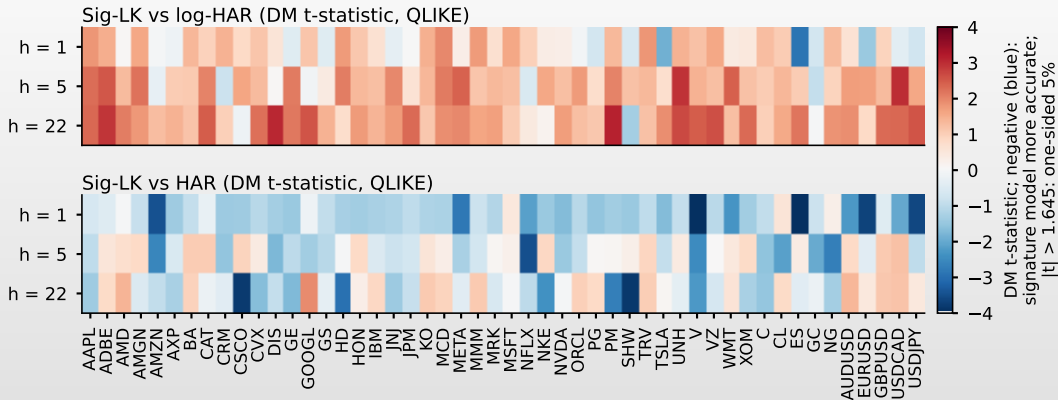
- $h = 1$: log-HAR, log-HAR-K and Sig-L are close (0.883, 0.885, 0.886)
 - ▶ Sig-LK is +4.5% relative to log-HAR: no gain from 36 extra features
- $h = 5, 22$: Sig-LK falls behind log-HAR (+28.0%, +92.9%)
 - ▶ Its medians (0.997, 0.968) are much better than its means: a few assets and days drive the loss
- HARQ and SHAR lose to HAR in levels here
 - ▶ The main design does not apply the BPQ filter (▶ **Appendix**: the filter rescues HARQ)
- First verdict: the method does not beat the simplest log model
 - ▶ A near-tie is not a proof of equal accuracy

Asset by Asset: Wins and Losses, Corrected

Comparison (QLIKE)	$h = 1$	$h = 5$	$h = 22$
Sig-LK vs log-HAR	2/0/0 5/0/0	0/0/0 15/0/2	0/0/0 27/1/16
Sig-L vs log-HAR	3/1/1 5/0/0	0/0/0 9/0/0	1/0/0 21/0/0
log-HAR-K vs log-HAR	3/0/0 3/0/0	3/0/0 1/0/0	5/0/0 6/0/0
Sig-LK vs Sig-L	2/0/0 5/0/0	0/0/0 5/0/0	2/0/0 10/0/0
Sig-LK vs HAR	12/5/6 0/0/0	7/1/1 0/0/0	8/2/4 1/0/0
log-HAR vs HAR	21/5/17 0/0/0	23/13/20 0/0/0	42/25/42 0/0/0

- Cell: significantly better on raw/Holm/BH assets | significantly worse on raw/Holm/BH assets; one-sided DM at 5%, family of 50 assets; Holm controls the FWER under any dependence, BH the FDR under independence or PRDS (positive regression dependence on a subset) [Benjamini & Yekutieli \(2001\)](#)
- H1 (signatures): Sig-L beats log-HAR on 3 of 50 assets at $h = 1$ raw, 1 after Holm; about 2.5 raw wins are expected by chance
- H2 (weights): log-HAR-K vs log-HAR and Sig-LK vs Sig-L: no Holm-significant difference at any horizon
- Against HAR at $h = 1$, Sig-LK wins on 12 assets, but log-HAR alone wins on 21: choose the benchmark that isolates the idea

DM Statistics for Every Asset and Horizon (1/2)

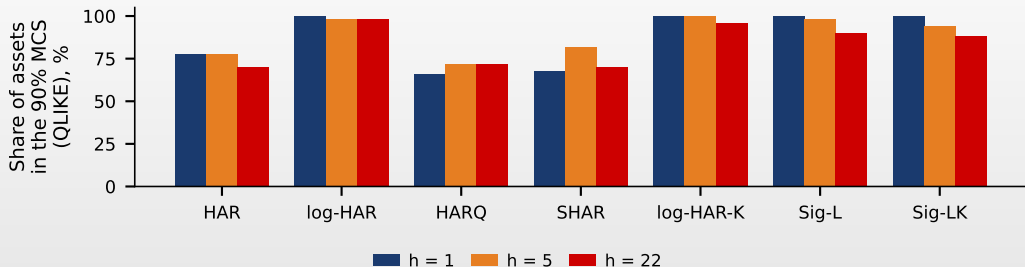


DM Statistics for Every Asset and Horizon (2/2)

- ▣ Diebold–Mariano t -statistics of the QLIKE differential; blue: Sig-LK more accurate, red: less accurate
- ▣ Against log-HAR (left): pale at $h = 1$ (blue exceptions ES, TSLA, EURUSD; five assets significantly red before correction), red at $h = 5, 22$
- ▣ Against HAR (right): blue at $h = 1$ because Sig-LK inherits the log target; one cell such as ES at $h = 1$ would make signatures look like a clear success

 MFM_ch20_case_study

Model Confidence Sets



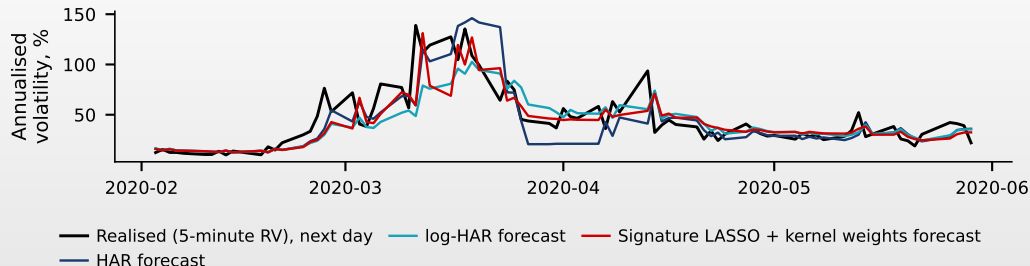
- Share of the 50 assets in which each model belongs to the 90% MCS under QLIKE: log-HAR 100%, 98%, 98%; Sig-LK 100%, 94%, 88% ($h = 1, 5, 22$)
- The MCS rarely separates the four log models; HAR, HARQ and SHAR are excluded for a fifth to a third of the assets: the level target is the clearest loser
- Under MSE almost every model stays in the set [Patton \(2011\)](#): small differences in average loss need very long samples [Pele & Mazurencu-Marinescu-Pele \(2026\)](#)

Panel Tests: Pooling the Evidence

Comparison: DM t (one-sided p)	$h = 1$	$h = 5$	$h = 22$
Sig-LK vs log-HAR	+1.45 (0.927)	+1.68 (0.953)	+1.80 (0.964)
Sig-L vs log-HAR	-1.91 (0.028)	+1.48 (0.931)	+1.51 (0.935)
log-HAR-K vs log-HAR	+1.89 (0.971)	+0.40 (0.654)	+1.92 (0.973)
Sig-LK vs Sig-L	+1.74 (0.959)	+0.98 (0.837)	-0.77 (0.221)
log-HAR vs HAR	-3.09 (<0.001)	-3.95 (<0.001)	-4.50 (<0.001)

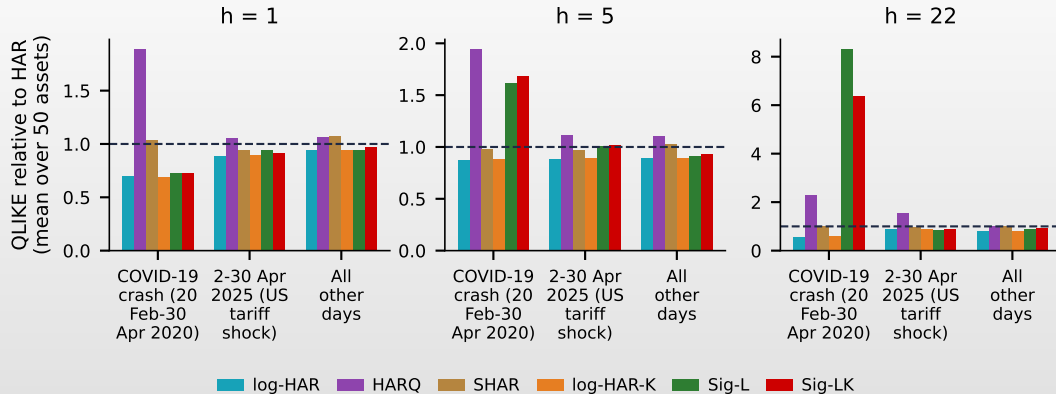
- Series: the mean QLIKE differential over all assets trading that day; negative t = first model better
- Pooled, only the log transform is clearly significant; Sig-L vs log-HAR at $h = 1$ is borderline ($p = 0.028$), driven by futures
 - For the 40 stocks alone the sign flips: $t = +1.72$ against Sig-L
- Conditional GW test, Sig-LK vs log-HAR: 0, 0 and 0 Holm rejections out of 50

Inside the COVID-19 Crash: JPM, Next Day



- Annualised volatility $\sqrt{252F_t}$ of the forecasts and $\sqrt{252RV_{t+1}}$ of the target, February–May 2020; all models lag the first jump
- Sig-LK reacts slightly faster than log-HAR in mid-March, when the path has already turned; after the peak both decay at similar speeds
- HAR in levels overshoots, then turns negative in late March: the positivity rule replaces it by the in-sample mean (the flat segment)

Stress Periods vs Normal Times (H4) (1/2)

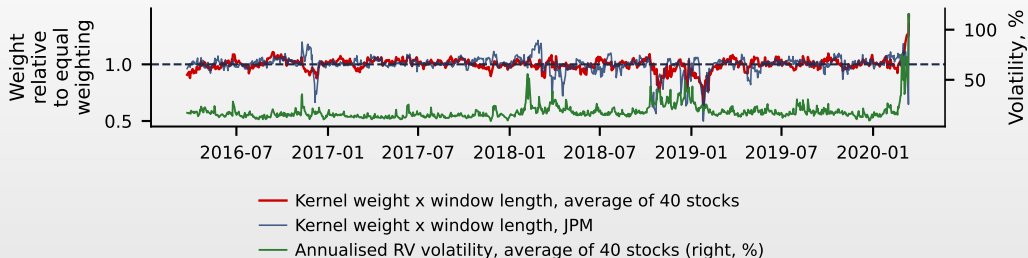


Stress Periods vs Normal Times (H4) (2/2)

- Mean over assets of QLIKE relative to HAR within each subperiod fixed before the evaluation
- $h = 1$, COVID-19 crash: log-HAR 0.70, Sig-LK 0.73, HARQ 1.89; 2–30 April 2020 (US tariff shock): 0.88 and 0.92; other days: 0.94 and 0.97
- $h = 22$, COVID-19 crash: Sig-LK 6.35 and Sig-L 8.30 times the loss of HAR: exactly where similar past windows should help, the signature models fail most; H4 not supported (descriptive, no formal interaction test)

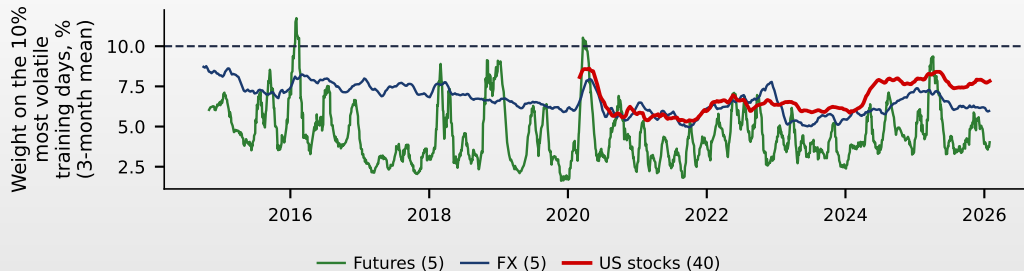
 MFM_ch20_case_study

Kernel Weights at the Origin 16 March 2020



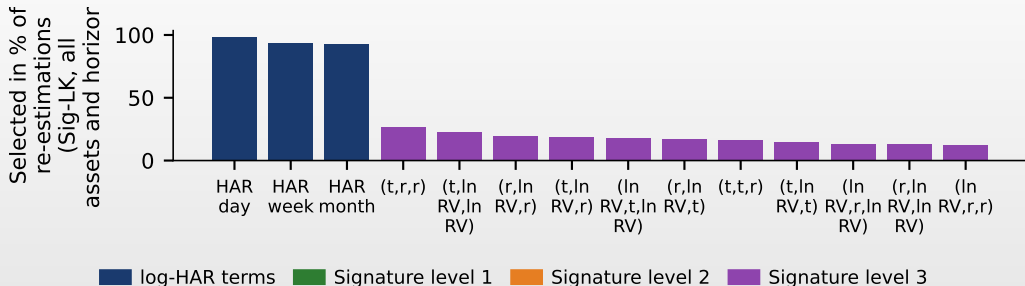
- Weights of Sig-LK ($h = 1$) at the origin 16 March 2020 over the training days (24 Mar 2016 – 13 Mar 2020), times the window length: 1 = equal weighting
- Nearly equal: between 0.61 and 1.43 times $1/n$, effective sample size 975 of 1000 (average over the 40 stocks); the highest weights go to the last days before the origin
- The late-2018 sell-off gets *less* weight; weights are almost uncorrelated with log volatility (+0.04): the translation-invariant signature sees shape, not level

Does the Kernel Ever Favour Stress?



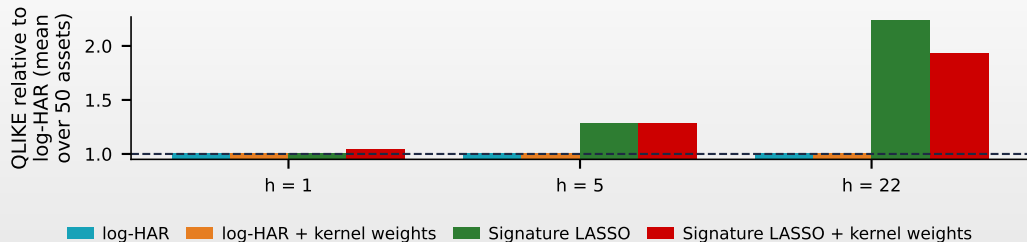
- Weight that Sig-LK ($h = 1$) puts on the 10% most volatile days of its training window, averaged by class, three-month mean
- Stocks: mean 6.7%, between 4.4% and 9.9% (maximum on 17 Mar 2020): always below the 10% of equal weighting; futures and FX show the same pattern
- The mechanism behind the null result for H2: the weights do not implement “learn from past crises”

What the LASSO Selects



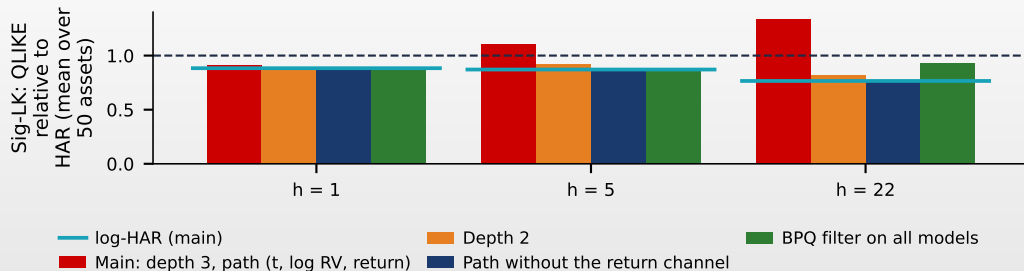
- Share of re-estimations in which a regressor is in the BIC-selected support of Sig-LK; t = time, $\ln RV = \ln RV$, r = cumulative return
- The log-HAR terms are kept in 99%, 94% and 93% of the re-estimations: the LASSO rediscovers HAR; average support 8.7 regressors (Sig-L), 6.7 (Sig-LK)
- The most frequent signature term, (t,r,r) (27%), and all frequent ones are of level 3: cubic in the increments, far outside the training range in a crash, amplified by $\exp(\cdot)$

The 2×2 Ablation



- Mean over the 50 assets of QLIKE relative to log-HAR: features (signature or not) crossed with weights (kernel or equal)
- Weights alone (log-HAR-K): 1.002, 1.002, 1.004: no effect
 - ▶ Average effective sample size 816 of 1000; features alone (Sig-L): 1.004, 1.285, 2.238
- Both (Sig-LK: 1.045, 1.280, 1.929) damp the one-month failure without removing it
 - ▶ They worsen $h = 1$ through crude oil (CL: 2.71); Gu et al. (Table 3) saw a kernel gain in six weekly values of one region

Robustness of Sig-LK



- Sig-LK relative to HAR (mean over 50 assets) in the main design, in two pre-registered variants and with the BPQ filter (the disclosed deviation); the line marks log-HAR
- Depth 2: 0.922 and 0.820 at $h=5, 22$ (main: 1.104, 1.334); path without returns: 0.878 and 0.770: the long-horizon failures come from level-3 terms with the return channel
- The stable variants only match log-HAR (0.871, 0.765): the null result for H1 does not depend on depth or channels, the instability does

By Asset Class

Sig-L relative to log-HAR (QLIKE)	$h = 1$	$h = 5$	$h = 22$
Stocks (40)	1.009	1.332	2.505
Futures (5)	0.965	1.084	1.206
FX (5)	0.996	1.111	1.134

- The only gain: futures at $h = 1$ (-3.5%) and, marginally, FX
 - Five futures and one horizon: a candidate for a new, pre-registered study, not a finding of this one
- Stocks, 80% of the panel, show no gain: the pooled result is mostly a stock result

Limitations and Open Directions

Limitations of this study

- One design out of many ($L = 22$, $N = 3$, three channels, BIC, a four-point grid for c): a null result for this design is not one for signatures in general
- Truncated kernel only; no external factors (implied volatility, macro news), central in Gu et al.
- Realized GARCH [Hansen, Huang & Shek \(2012\)](#) and machine-learning benchmarks ([Chapter 13](#)) are outside the comparison; stocks are evaluated over six years only

Open directions

- A level-aware kernel: basepoint augmentation or the level of $\ln RV$ in the distance (Seminar 20, C1)
- Signatures of the intraday path (5-minute returns); the PDE kernel [Salvi et al. \(2021\)](#); log-signatures and bounded (tanh) channels against explosive level-3 terms [Morrill et al. \(2020\)](#)
- Cross-asset paths, where Lévy areas measure lead-lag across markets; futures at $h = 1$ on a longer panel, with this design fixed in advance

An Open Question

When does a method transfer across domains?

- Methods travel faster than evidence: a KDD paper on freight rates becomes a candidate for volatility, credit, energy
 - ▶ Which properties of the source problem made the method work?
 - ▶ Does the target share them?
- Finance adds a hard filter: 164 teams on the same data disagree more than their standard errors suggest [Menkveld et al. \(2024\)](#)
- Formal version: a transfer succeeds if the gain over the strongest simple benchmark survives a pre-registered set $\mathcal{S}$ of designs, assets and horizons, with the family corrected
 - ▶ In this chapter: $\mathcal{S} = 50 \text{ assets} \times 3 \text{ horizons} \times 4 \text{ variants}$; Holm/BH were applied per comparison and horizon (families of 50 assets), variants reported descriptively; no joint correction over all 600 cells; the gain did not survive

The Discovery Loop Applied to This Chapter

- **Read:** an assistant lists every equation, hyperparameter and benchmark of the paper and marks what is not disclosed; each item is checked against the PDF
 - ▶ **Map:** for each component, does it transfer, what must change, which benchmark isolates it
- **Pre-register:** hypotheses, grid, tests and correction fixed in a dated document before the first result; the assistant drafts, the team decides every number
- **Build and test the pieces:** signatures against identities (level 1 = increment, Chen on a split path), not against another library
- **Run the grid, then attack it**
 - ▶ which conclusions survive Holm over 50 assets?
 - ▶ which single asset would make a misleading headline?
- **Explain the null:** here flat weights (translation invariance) and explosive level-3 terms (exponential link), both checked with data; the full prompts and the researcher's checklist are in the [▶ Appendix](#)

Mini-Case: Three Headlines From the Same Runs

- “Signatures beat HAR on 12 of 50 assets” (one-day horizon, raw DM)
 - ▶ True, but log-HAR alone beats HAR on 21: the gain belongs to the log transform
- “Signatures improve E-mini S&P 500 volatility forecasts” (ES, $h = 1$, DM $t = -2.91$ against log-HAR)
 - ▶ One cell out of 150 asset-horizon cells: after Holm, 0 of 50 at $h = 1$ and none at $h = 5, 22$
- The headline of the pre-registered evaluation: “no gain over log-HAR; worse at longer horizons; kernel weights inert”
 - ▶ An AI assistant asked to “find where the method works” will produce the first two; the plan protects the third

Your Turn

- Pick one open direction of this chapter (level-aware kernel, intraday signatures, log-signatures, futures only)
 - ▶ Write H_0/H_1 , the benchmark that isolates it, and the result that would make you drop it
- Use an assistant to list what could go wrong in your design, then write one unit test per item
- Fix the pre-registration in a dated document before the first out-of-sample number; run the grid; report the distribution over $\mathcal{S}$
- Log every AI error you catch in `AI_ERRORS.md` (Seminar 20, C2 gives typical ones)

Key Takeaways

- Signatures turn a window into a feature vector with a theory behind it: Chen's identity, uniqueness, universality
- Transferred to realised variance, the method of Gu et al. does not beat log-HAR
 - ▶ H1 (signatures add information): not supported, no gain at one day, a loss at a week and a month
 - ▶ H2 (kernel weights help): not supported, weights are almost flat and down-weight past stress; H3: only HAR in levels is beaten, through the log target; H4 (gains in stress): not supported
- The translation-invariant kernel sees shape, not level; level-3 terms explode in crashes
- The strongest effect in the study is the oldest one: model the log of RV
- A pre-registered null result, with ablations and a mechanism, is a finding

Further Study

Reading

- ▣ Signatures: Chevyrev & Kormilitzin (2016); Lyons & McLeod (2022); realised measures: Franke, Härdle & Hafner (2019) and Chapter 9

Advanced courses on quantinar.com

- ▣  Statistics of Financial Markets
- ▣  Applied Time Series Analysis with Python
- ▣  Machine Learning in Financial Risk

Code and data

- ▣ Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_20; data: VOLARE, free for research at volare.unime.it

Appendix: Features and Scaling

- Regressors at day τ for the signature models: 3 log-HAR terms + 39 signature terms
 - ▶ $\xi_\tau = [\ln \text{RV}_\tau, \ln \text{RV}_\tau^{(w)}, \ln \text{RV}_\tau^{(m)}, \text{Sig}^3(X_{\tau-21:\tau})]$, path $X = (\text{time}, \ln \text{RV}, \text{cumulative } r)$, 22 points
- Each non-time channel q is multiplied by $\kappa_q = 1/(s_q\sqrt{L-1})$, so its typical total move over the window is about 1
 - ▶ s_q : standard deviation of the daily increments of channel q in the training window up to the origin t ; $L-1 = 21$ increments per window
 - ▶ Time is not rescaled (constant increments, $s = 0$): it runs from 0 to 1; a word $(i_1..i_k)$ is rescaled by $\prod_j \kappa_{i_j}$, so signatures are computed once
- Before the LASSO every column is standardised with weighted moments of the training window; constant columns (words of time only) are dropped
- Raw features at τ use data up to τ ; scales and standardisation use data up to the origin t : the no-look-ahead cutoff is t

Appendix: Estimation: Choosing λ Without Look-Ahead

- Inside each training window: the whole weighted LASSO path by LARS (least-angle regression)
 - ▶ For every support on the path: weighted OLS refit (step 2 of Gu et al.)
- Pick the support that minimises a BIC-type criterion (Bayesian information criterion) with the Kish effective sample size, a heuristic of this design
 - ▶ $\text{BIC}(S) = n_{\text{eff}} \ln \hat{s}_S^2 + (|S| + 1) \ln n_{\text{eff}}$, $n_{\text{eff}} = 1 / \sum_{\tau} w_{\tau}^2$
 - ▶ S : a support (set of kept regressors), $|S|$: its size, $+1$ for the intercept; $\hat{s}_S^2$: weighted residual variance of the refit on S ; first term: fit, second: penalty per parameter
- Why not cross-validation: 50 assets $\times$ 3 horizons $\times$ hundreds of re-estimations; BIC uses only the window
 - ▶ Kish size corrects for unequal weights, not for serial dependence (overlapping targets, $h > 1$): not a likelihood-based BIC for these data

Appendix: Kernel Temperatures Chosen on the Validation Block

- ▣ Chosen c for Sig-LK ($h = 1, 5, 22$)
 - ▶ stocks 0.5, 0.5, 0.5; futures 2.0, 1.0, 0.5; FX 0.5, 2.0, 0.5
- ▣ Chosen c for log-HAR-K ($h = 1, 5, 22$)
 - ▶ stocks 0.5, 0.5, 1.0; futures 2.0, 1.0, 4.0; FX 4.0, 2.0, 4.0
- ▣ Grid $c \in \{0.5, 1, 2, 4\}$; the smallest temperature (the most nearly equal weights) wins for the stocks at every horizon; for futures and FX the choice changes with the horizon

Appendix: What Was Evaluated

Class	Evaluation starts	Ends	Days per asset (median)
Stocks	27 Jan 2020	29 Jan 2026	1 499
Futures	25 Sep 2014	29 Jan 2026	2 919
FX	27 Aug 2014	29 Jan 2026	2 952

- 265 929 out-of-sample forecasts per model ($50 \text{ assets} \times 3 \text{ horizons}$)
- Stocks start in 2015, so their evaluation begins in January 2020: the COVID-19 crash is in every stock sample
 - Futures and FX start in 2009 and are evaluated from 2014

Appendix: The Paper's Metric on VOLARE

Median relative error $ RV - F /RV$ (%)	$h = 1$	$h = 5$	$h = 22$
HAR	57.2	48.9	51.7
log-HAR	50.4	41.9	38.7
HARQ	55.0	49.0	50.7
SHAR	57.9	49.2	51.8
log-HAR-K	49.7	41.0	38.7
Sig-L	50.1	42.6	42.6
Sig-LK	50.6	43.6	43.7

- Median over assets of the mean relative error: about 40–60%, against 1–2% for freight rates in Gu et al.
- Daily RV is a noisy, heavy-tailed target: the evaluated models have much larger relative errors than those reported for freight rates, without establishing a lower bound on attainable error
 - ▶ Sig-LK forecasts above ten times the in-sample maximum: 8, 29, 36 ($h = 1, 5, 22$), on 10 assets at $h = 5$; log-HAR: none

Appendix: The Insanity Filter, Revisited

QLIKE relative to HAR, BPQ filter on all models	$h = 1$	$h = 5$	$h = 22$
HAR	1.000	1.000	1.000
log-HAR	0.763	0.654	0.778
HARQ	0.995	0.912	1.014
SHAR	1.014	1.003	1.000
log-HAR-K	0.772	0.659	0.779
Sig-L	0.871	0.819	0.904
Sig-LK	0.890	0.867	0.930

- The filter rescues HARQ relative to HAR (0.995 at $h = 1$ instead of 1.140), as BPQ intended
- Sig-LK relative to log-HAR, both filtered: 1.238, 1.424, 1.209 (unfiltered 1.045, 1.280, 1.929): the gap widens at $h = 1, 5$ and narrows at $h = 22$
- Filtered forecasts, $h = 1$: HAR 180, HARQ 140, Sig-LK 143, log-HAR 29 (of 89 893): the filter hits the signature models far more often than log-HAR

Appendix: What the Researcher Must Check

- **The method is the paper's:** kernel, weights, refit, as in the cited equations
 - ▶ Test: every design slide cites an equation, table or section of the paper; undisclosed choices are labelled as choices of the study
- **No look-ahead:** λ , scales, c and features use only data available at the origin
 - ▶ Test: change every value after the origin (for example, multiply future RV by 10); the forecast made at the origin must not change
- **The benchmark isolates the idea**
 - ▶ Test: the 2×2 ablation is reported, not only the full model against HAR
- **Losses and tests are the right way round**
 - ▶ Test: $QLIKE(RV, RV) = 0$ and $QLIKE(RV, F) > 0$ otherwise; the DM variance is HAC with bandwidth at least h

Appendix: The Discovery Loop, Prompts (1/2)

- **Read:** extract exactly what the paper did
 - ▶ “From Gu et al. (2024, KDD), list every equation of the estimator, every hyperparameter and its reported value, and the benchmark of each table. Mark what is not disclosed.”
 - ▶ Check each item against the PDF: equation numbers, table numbers, the kernel actually used
- **Map:** what transfers, what must be adapted
 - ▶ “Our target is daily realised variance. For each component of the method, say whether it transfers, what must change, and which benchmark isolates it.”
- **Pre-register:** hypotheses, grid, tests, correction, fixed in a dated document before the first result
 - ▶ The assistant drafts; the team decides every number and signs the dated pre-registration

Appendix: The Discovery Loop, Prompts (2/2)

- **Build and test the pieces:** signatures against identities, not against another library
 - ▶ “Write unit tests for a depth-3 signature: level 1 equals the increment, the symmetric part of level 2 equals half the outer product, Chen's identity on a split path.”
- **Run the grid, then attack it**
 - ▶ “Here are all 600 runs (50 assets $\times$ 3 horizons $\times$ 4 specifications: main, depth 2, no return channel, BPQ filter). Which conclusions survive Holm over 50 assets? Which single asset or subperiod would make a misleading headline?”
- **Explain the null:** find the mechanism, not an excuse
 - ▶ Here: flat weights (translation invariance) and explosive level-3 terms (exponential link), both checked with data

References (1/3)

- ▣ Barndorff-Nielsen, O.E., Hansen, P.R., Lunde, A., Shephard, N. (2008). [Designing realized kernels to measure the ex post variation of equity prices in the presence of noise](#). *Econometrica*, 76(6), 1481–1536.
- ▣ Barndorff-Nielsen, O.E., Kinnebrock, S., Shephard, N. (2010). [Measuring downside risk: realised semivariance](#). In *Volatility and Time Series Econometrics*, Oxford University Press, 117–136.
- ▣ Barndorff-Nielsen, O.E., Shephard, N. (2002). [Econometric analysis of realized volatility and its use in estimating stochastic volatility models](#). *Journal of the Royal Statistical Society B*, 64(2), 253–280.
- ▣ Belloni, A., Chernozhukov, V. (2013). [Least squares after model selection in high-dimensional sparse models](#). *Bernoulli*, 19(2), 521–547.
- ▣ Benjamini, Y., Hochberg, Y. (1995). [Controlling the false discovery rate: a practical and powerful approach to multiple testing](#). *Journal of the Royal Statistical Society B*, 57(1), 289–300.
- ▣ Benjamini, Y., Yekutieli, D. (2001). [The control of the false discovery rate in multiple testing under dependency](#). *The Annals of Statistics*, 29(4), 1165–1188.
- ▣ Bollerslev, T., Patton, A.J., Quaedvlieg, R. (2016). [Exploiting the errors: a simple approach for improved volatility forecasting](#). *Journal of Econometrics*, 192(1), 1–18.
- ▣ Chen, K.-T. (1957). [Integration of paths, geometric invariants and a generalized Baker–Hausdorff formula](#). *Annals of Mathematics*, 65(1), 163–178.
- ▣ Chevyrev, I., Kormilitzin, A. (2016). [A primer on the signature method in machine learning](#). [arXiv:1603.03788](#).
- ▣ Cipollini, F., Cruciani, G., Gallo, G.M., Insana, A., Otranto, E., Spagnolo, F. (2026). [VOLatility Archive for Realized Estimates \(VOLARE\)](#). [arXiv:2602.19732](#).
- ▣ Corsi, F. (2009). [A simple approximate long-memory model of realized volatility](#). *Journal of Financial Econometrics*, 7(2), 174–196.

References (2/3)

- Cuchiero, C., Gazzani, G., Svaluto-Ferro, S. (2023). [Signature-based models: theory and calibration](#). *SIAM Journal on Financial Mathematics*, 14(3), 910–957.
- Diebold, F.X., Mariano, R.S. (1995). [Comparing predictive accuracy](#). *Journal of Business & Economic Statistics*, 13(3), 253–263.
- Franke, J., Härdle, W.K., Hafner, C.M. (2019). [Statistics of Financial Markets: An Introduction](#), 5th ed. Springer.
- Giacomini, R., White, H. (2006). [Tests of conditional predictive ability](#). *Econometrica*, 74(6), 1545–1578.
- Gu, H., Guo, X., Jacobs, T.L., Kaminsky, P., Li, X. (2024). [Transportation marketplace rate forecast using signature transform](#). *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, 4997–5005; [arXiv:2401.04857](#).
- Hambly, B., Lyons, T. (2010). [Uniqueness for the signature of a path of bounded variation and the reduced path group](#). *Annals of Mathematics*, 171(1), 109–167.
- Hansen, P.R., Huang, Z., Shek, H.H. (2012). [Realized GARCH: a joint model for returns and realized measures of volatility](#). *Journal of Applied Econometrics*, 27(6), 877–906.
- Hansen, P.R., Lunde, A., Nason, J.M. (2011). [The model confidence set](#). *Econometrica*, 79(2), 453–497.
- Király, F.J., Oberhauser, H. (2019). [Kernels for sequentially ordered data](#). *Journal of Machine Learning Research*, 20(31), 1–45.
- Levin, D., Lyons, T., Ni, H. (2013). [Learning from the past, predicting the statistics for the future, learning an evolving system](#). [arXiv:1309.0260](#).
- Liu, L.Y., Patton, A.J., Sheppard, K. (2015). [Does anything beat 5-minute RV? A comparison of realized measures across multiple asset classes](#). *Journal of Econometrics*, 187(1), 293–311.

References (3/3)

- Lyons, T. (1998). [Differential equations driven by rough signals](#). *Revista Matemática Iberoamericana*, 14(2), 215–310.
- Lyons, T., McLeod, A.D. (2022). [Signature methods in machine learning](#). arXiv:2206.14674.
- Menkveld, A.J., Dreber, A., Holzmeister, F., Huber, J., Johannesson, M., Kirchler, M., et al. (2024). [Nonstandard errors](#). *Journal of Finance*, 79(3), 2339–2390.
- Morrill, J., Fermanian, A., Kidger, P., Lyons, T. (2020). [A generalised signature method for multivariate time series feature extraction](#). arXiv:2006.00873.
- Patton, A.J. (2011). [Volatility forecast comparison using imperfect volatility proxies](#). *Journal of Econometrics*, 160(1), 246–256.
- Patton, A.J., Sheppard, K. (2015). [Good volatility, bad volatility: signed jumps and the persistence of volatility](#). *Review of Economics and Statistics*, 97(3), 683–697.
- Pele, D.T., Mazurencu-Marinescu-Pele, M. (2026). [Finite-sample precision limits for expected shortfall forecast comparisons](#). *Mathematics*, 14(13), 2316.
- Salvi, C., Cass, T., Foster, J., Lyons, T., Yang, W. (2021). [The signature kernel is the solution of a Goursat PDE](#). *SIAM Journal on Mathematics of Data Science*, 3(3), 873–899.
- Tibshirani, R. (1996). [Regression shrinkage and selection via the lasso](#). *Journal of the Royal Statistical Society B*, 58(1), 267–288.
- Zou, H. (2006). [The adaptive lasso and its oracle properties](#). *Journal of the American Statistical Association*, 101(476), 1418–1429.